### (iii) When deviations are taken from an Assumed mean

When actual means are in fractions, say the actual means of $X$ and $Y$ series are 20.167 and 29.23, the calculation of correlation by the method discussed above would involve too many calculations and would take a lot of time. In such cases we make use of the assumed mean method for finding out correlation. When deviations are taken from an assumed mean the following formula is applicable:

$$r = \frac{N\Sigma\, dx\, dy - (\Sigma dx)(\Sigma dy)}{\sqrt{N\Sigma dx^2 - (\Sigma dx)^2} \times \sqrt{N\Sigma dy^2 - (\Sigma dy)^2}}$$

Where $dx = X - A$ and $dy = Y - B$. Here $A$ and $B$ are assumed mean

**NOTE**

While applying assumed mean method, any value can be taken as the assumed mean and the answer will be the same. However, the nearer the assumed mean to the actual mean, the lesser will be the calculations.

### Steps to solve the problems:

(i) Take the deviations of $X$ series from an assumed mean, denote these deviations by $dx$ and obtain the total that is $\Sigma dx$.

(ii) Take the deviations of $Y$ series from an assumed mean, denote these deviations by d$y$ and obtain the total that is $\Sigma dy$.

(iii) Square $dx$ and obtain the total $\Sigma dx^2$.

(iv) Square $dy$ and obtain the total $\Sigma dy^2$.

(v) Multiply $dx$ and d$y$ and obtain the total $\Sigma dx\, dy$.

(vi) Substitute the values of $\Sigma\, dxdy$, $\Sigma\, dx$, $\Sigma\, dy$, $\Sigma\, dx^2$ and $\Sigma\, dy^2$ in the formula given above.

### Example 9.3

Find out the coefficient of correlation in the following case and interpret.

| Height of father (in inches) | 65 | 66 | 67 | 67 | 68 | 69 | 71 | 73 |
|---|---|---|---|---|---|---|---|---|
| Height of son (in inches) | 67 | 68 | 64 | 68 | 72 | 70 | 69 | 70 |

***Solution:***

Let us consider Height of father (in inches) is represented as $X$ and Height of son (in inches) is represented as $Y$.

| $X$ | $dx = (X-67)$ | $dx^2$ |
|---|---|---|
| 65 | -2 | 4 |
| 66 | -1 | 1 |
| 67 | 0 | 0 |
| 67 | 0 | 0 |
| 68 | 1 | 1 |
| 69 | 2 | 4 |
| 71 | 4 | 16 |
| 73 | 6 | 36 |
| $\Sigma\ X = 546$ | $\Sigma\ dx = 10$ | $\Sigma\ dx^2 = 62$ |

| $Y$ | $dy = (Y-68)$ | $dy^2$ | $dxdy$ |
|---|---|---|---|
| 67 | –1 | 1 | 2 |
| 68 | 0 | 0 | 0 |
| 64 | -4 | 16 | 0 |
| 68 | 0 | 0 | 0 |
| 72 | 4 | 16 | 4 |
| 70 | 2 | 4 | 4 |
| 69 | 1 | 1 | 4 |
| 70 | 2 | 4 | 12 |
| $\Sigma\ Y = 548$ | $\Sigma\ dy = 4$ | $\Sigma\ dy^2 = 42$ | $\Sigma\ dxdy = 26$ |

Table 9.3

$$r = \frac{N\,\Sigma dx\, dy - (\Sigma dx)(\Sigma dy)}{\sqrt{N\Sigma dx^2 - (\Sigma dx)^2} \times \sqrt{N\Sigma dy^2 - (\Sigma dy)^2}}$$

Here $\Sigma dx = 10$, $\Sigma dx^2 = 62$, $\Sigma dy = 4$, $\Sigma dy^2 = 42$ and $\Sigma dxdy = 26$

$$r = \frac{(8 \times 26) - (10 \times 4)}{\sqrt{(8 \times 62) - (10)^2} \times \sqrt{(8 \times 42) - (4)^2}}$$



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